

KHABAROVSK, Russian Federation

WMO: 317350

Lat: **48.528N**Lon: **135.188E**Elev: **249**StdP: **14.56**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-18.6	-29.0	1.1	-20.4	-25.6	1.3	-17.4	27.4	7.3	24.5	-1.0	4.0	240	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	86.4	72.6	83.8	71.2	80.7	69.7	75.6	82.0	73.7	80.1	72.1	78.3	8.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	126.3	78.3	71.7	118.3	76.9	69.9	110.9	75.6	39.3	81.4	37.6	79.5	36.1	78.4	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.4	19.1	-27.8	91.3	5.1	3.7	-31.5	93.9	-34.5	96.1	-37.4	98.1	-41.1	100.8		
(5)			-28.0	78.4	5.1	1.9	-31.7	79.8	-34.6	80.9	-37.5	82.0	-41.1	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	-4.7	3.5	20.0	40.2	54.8	64.6	71.0	68.1	57.3	41.2	18.6	-0.7
	DBStd	27.87	8.59	8.66	9.87	8.26	7.44	6.48	5.09	5.05	6.58	7.89	11.21	8.77
	HDD50	7097	1697	1303	930	313	37	0	0	0	14	289	942	1572
	HDD65	10879	2162	1723	1395	744	325	85	10	26	242	738	1392	2037
	CDD50	2105	0	0	0	20	186	439	650	562	232	16	0	0
	CDD65	411	0	0	0	0	9	73	196	123	10	0	0	0
	CDH74	3134	0	0	0	9	143	684	1478	746	73	1	0	0
	CDH80	770	0	0	0	1	22	181	434	127	5	0	0	0
	WSAvg	8.3	7.3	8.3	9.3	9.2	8.8	7.4	6.5	6.8	7.7	9.5	10.4	9.0
	PrecAvg	26.5	0.5	0.4	0.7	1.6	2.4	3.0	5.3	5.5	3.5	1.9	0.9	0.6
(15) Precipitation	PrecMax	43.3	1.4	1.3	1.8	5.1	5.7	6.5	11.9	17.1	9.2	5.0	2.8	1.7
	PrecMin	14.9	0.0	0.0	0.0	0.0	0.6	0.1	0.3	1.0	0.6	0.0	0.1	0.1
	PrecStd	5.4	0.4	0.3	0.4	0.9	1.1	1.6	3.0	3.1	2.0	1.3	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.4	33.8	49.8	73.6	82.7	89.4	91.8	86.7	79.9	69.9	48.5	28.7
		MCWB	19.5	28.6	39.0	54.3	63.1	71.4	73.8	75.9	67.5	56.3	40.7	26.2
	2%	DB	16.0	27.3	43.1	66.3	77.6	84.6	88.0	83.4	75.6	62.7	42.9	21.1
		MCWB	14.7	23.5	35.1	49.8	59.1	69.2	73.5	73.3	64.9	52.0	37.8	19.2
	5%	DB	12.0	21.6	38.7	59.4	73.7	81.0	84.6	80.7	73.0	57.5	38.8	15.8
		MCWB	10.7	18.8	32.7	46.2	57.3	67.5	72.4	71.4	63.1	48.6	34.4	14.5
	10%	DB	8.4	17.3	34.8	54.8	69.7	77.6	81.9	78.5	69.5	53.7	34.1	12.1
		MCWB	7.1	15.2	30.1	44.1	55.9	66.2	71.5	69.8	61.2	46.3	31.1	11.0
	0.4%	WB	19.6	28.7	39.6	55.2	65.4	74.9	77.9	78.8	69.9	57.8	42.9	26.5
		MCDB	21.1	33.0	47.3	71.2	77.8	84.6	85.2	84.1	76.5	66.0	47.5	28.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	14.7	23.3	36.2	51.1	61.8	71.4	75.6	75.9	67.4	53.6	38.0	19.2
		MCDB	15.9	27.0	41.9	63.2	73.5	80.6	83.2	80.0	73.0	61.2	41.9	20.8
	5%	WB	10.6	19.2	33.0	47.8	59.6	69.3	74.2	73.6	65.0	50.3	34.4	14.2
		MCDB	11.7	21.7	37.6	58.5	69.8	78.6	81.5	78.3	70.5	55.6	37.7	15.5
	10%	WB	6.9	15.0	30.3	44.7	57.4	67.4	72.7	71.7	62.4	47.1	31.4	10.8
		MCDB	8.0	16.9	34.6	53.1	67.4	75.5	79.6	76.6	67.6	52.6	34.1	11.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	17.4	17.1	18.7	20.5	19.0	17.0	15.9	18.3	17.2	13.7	13.2
		MCDBR	16.4	19.8	19.2	26.5	27.7	24.0	21.8	19.6	22.5	23.3	17.3	15.0
		MCWBR	15.1	17.0	13.1	14.6	13.3	11.2	9.9	10.5	12.4	14.3	13.2	13.9
		MCDBR	16.3	19.6	18.0	24.9	23.6	21.1	18.5	16.5	19.2	21.3	15.8	14.6
	5% WB	MCWBR	15.1	17.0	13.1	14.7	13.0	11.0	9.8	10.1	11.8	14.7	12.8	13.7
(40) Clear-Sky Solar Irradiance	taub		0.287	0.332	0.406	0.532	0.559	0.515	0.503	0.448	0.413	0.438	0.370	0.301
	taud		2.089	1.954	1.833	1.746	1.776	1.986	2.063	2.224	2.224	1.979	2.020	2.062
	Ebn at Noon		237	249	247	228	228	241	242	251	247	207	196	210
	Edh at Noon		28	41	55	66	67	55	50	41	37	40	30	25
(44) All-Sky Solar Radiation	RadAvg		592	989	1449	1551	1698	1804	1672	1477	1217	823	537	457
	RadStd		24	32	90	148	142	219	145	95	92	57	36	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page